

Biruk Ambaye

Uppsala, Sweden · +46764502691 · akibiruk39@gmail.com
github.com/Biruk-aki · biruk-embedded.vercel.app

SUMMARY

Embedded Systems Master's student at Uppsala University specializing in low-level firmware development, RTOS architectures, and deterministic control systems. Experienced in implementing multi-threaded applications using Zephyr RTOS on ARM Cortex-M33 and ESP32-S3 platforms, developing ADC/PWM control loops with digital signal filtering, and executing automated hardware-in-the-loop (HIL) test suites. Strong background in sensor data acquisition (I2C, SPI, UART), RF communication, and real-time scheduling.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, JavaScript (React.js)
- **Embedded & RTOS:** Zephyr RTOS, FreeRTOS, ESP32-S3, ARM (Cortex-M33 / RP2350), Raspberry Pi Pico (RP2040), AVR (ATmega)
- **Protocols & Interfacing:** I2C, SPI, UART, PWM, ADC, CAN bus, USB Serial
- **RF & Wireless:** Dyadic backscatter, 2-FSK modulation, sub-GHz (NRF52840, CC1352, CC2500), BER/PER analysis
- **Tools & Frameworks:** PySerial, pytest, Logic Analyzers, Oscilloscopes, GDB, Git, Linux/POSIX, CMake, West

EMBEDDED PROJECTS

ESP32-S3 Multi-Threaded Smart Controller 2026

Zephyr RTOS, ESP32-S3, Hardware PWM, ADC, GPIO Interrupts, Workqueues

- Designed multi-threaded firmware on ESP32-S3 featuring an 8-sample moving average DSP filter and a 3-State Finite State Machine (Manual Dimmer, Breathe, Emergency Strobe).
- Implemented thread synchronization between low-priority ADC sampling (20 Hz) and high-priority FSM actuation loops.
- Developed edge-triggered GPIO interrupt handling with delayable workqueues for hardware button debouncing.

Dyadic Backscatter Communication — Baud Rate Optimization 2026

Research Project | Uppsala University | NRF52840, Raspberry Pi Pico, CC1352, CC2500

- Built an experimental backscatter testbed with NRF52840 as carrier, Raspberry Pi Pico with dual CC2500 antennas as passive 2-FSK tag, and CC1352 as receiver.
- Evaluated baud rate reduction (100 kBaud → 50 kBaud) to improve SNR and link reliability without raising transmit power.
- Authored automated Python scripts to parse 15 measurement logs, evaluate Bit Error Rate (BER)/Packet Error Rate (PER), and validate hardware sync-word rejection.

Multi-Node Health Monitoring System 2026

Zephyr RTOS, Raspberry Pi Pico 2, KY-039, MIKROE-3102 SpO2, BME680, ADC, I2C, UART

- Built a two-node monitoring setup running Zephyr RTOS to acquire heart rate (ADC), SpO2 (I2C), and ambient temperature (I2C).
- Implemented inter-node UART streaming to a base station with real-time serial visualization on a host PC.

Environmental Monitoring & Hardware Alert System 2026

Zephyr RTOS | RP2350 (ARM Cortex-M33)

- Developed BME680 sensor firmware managing MOX heater plate cycling to evaluate Gas Resistance and IAQ scores.
- Integrated Devicetree overlays (.overlay) and Kconfig configurations for sensor APIs and software PWM alert drivers.

Deterministic Real-Time Scheduler 2025

Embedded C | AVR (ATmega)

- Developed a priority-based deterministic scheduler for AVR microcontrollers to manage periodic real-time tasks.
- Performed Worst-Case Execution Time (WCET) analysis to guarantee hard real-time deadline compliance.

EDUCATION

Uppsala University	2024 – Present
Master of Science in Embedded Systems	Sweden
Focus: Real-Time Systems, Microcontroller Interfacing, Hardware-Software Co-Design	
Mekelle University	2017 – 2024
Bachelor of Science in Computer Science	Ethiopia

EXPERIENCE

Aynalem Electronics	2024 – 2025
Junior Frontend Web Developer	Ethiopia
• Developed modular React.js interfaces adhering to structured code review and quality standards. • Optimized front-end performance and state management logic across web platforms.	
Proxima Technologies	2023 – 2024
Frontend Web Developer	Ethiopia
• Collaborated in agile Scrum sprints utilizing Git version control for continuous integration and documentation.	

LANGUAGES

English: Professional working proficiency · **Swedish:** Basic (Actively Studying) · **Amharic / Tigrigna:** Native